Tony Hsu Tai

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Education

Northeastern University Sep 2024 - May 2027
M.S. in Computer Science | ***GPA: 3.92/4.00*** *Boston, MA*

- Coursework: Data Structures and Algorithms, Foundations of AI, Scalable Distributed Systems, Database Management Systems

Northeastern University Sep 2016 - May 2021
Accelerated B.S./M.S. Program in Industrial Engineering | ***GPA: 3.75/4.00*** *Boston, MA*

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Nest.JS, Next.js, Express.js, FastAPI, Tailwind CSS

Database Management: PostgreSQL, MySQL, MongoDB

Tools & Platforms: AWS (S3, EKS, SQS), Git/Github, Azure DevOps, Docker, Postman, Claude Code

Professional Experience

Software Developer Intern May 2026 – Present
Sentinel Group *Wakefield, MA*

- Built an **Airflow ETL pipeline** that migrates payroll and deposit configurations from an Oracle CRM into our new payroll applications, replacing manual data entry for 1,700+ clients.
- Enhanced enterprise search engine with **grounding responses in internal source code through an Azure DevOps repository crawler** that indexes 25,000+ source files while preserving access permissions and minimizing redundant indexing.
- Reduced local development setup time by 93% by replacing LocalStack with license-free Floci for SQS/S3 emulation and **automating configuration of 12 services and AWS resources** through a single setup script.
- Automated pay-period reporting by replacing manual Excel building process with automatically generated, formatted spreadsheets delivered to users, **built on an asynchronous SQS queue** with a dead-letter queue for reliable processing.
- Expanded PDF reporting capabilities by designing a custom solution that lets business users **embed data visualizations within the existing template builder**, overcoming the builder's technical challenges.

Operations Manager Sep 2021 – Jun 2024
Boston Microgreens *Boston, MA*

- Built an **Excel-based scheduling system** that automated seeding, germination, and harvest planning for 50+ weekly orders, becoming the farm's single source of truth for reliable crop and order data that remains in daily use after 4+ years.
- Designed a **germination tracking system using relational databases** and computed fields to track crop stage transitions, automatically flagging what crops to check and move to irrigation area, reducing task time by 63%.
- Built a **per-crop cost model** that combined each variety's raw material costs with operation costs allocated by growth time and resource usage, revealing 11% of varieties were not generating profit and driving strategic adjustments to improve margins.

Personal Projects

AI Financial Analysis Platform | *TypeScript, Node.js, PostgreSQL, React, Gemini API* Dec 2025 – June 2026

- Built an **asynchronous PDF statement ingestion pipeline** in TypeScript and PostgreSQL that extracts account and transaction data with layered regex and tracks each upload through a status lifecycle.
- Engineered a **three-tier transaction classifier** using 890+ merchant aliases, 55 regex rules, and a **Gemini fallback** that sends only unmatched transactions in one batched call per statement.
- Designed an **LLM finance chat** that turns questions into **user-scoped, parameterized SQL** through schema-aware prompting and fills answer templates by querying data, so all figures come from the database.

Agricultural Operations System | *Python, TypeScript, MySQL, React, FastAPI* Nov 2025 – Dec 2025

- Designed a full-stack farm operations platform with React, FastAPI and MySQL, **replacing spreadsheet workflows with a relational data model** that enforces integrity through foreign-key constraints and stored-procedure validation.
- Built a **SQL view and stored procedures that work backward from customer delivery dates** to compute planting, germination dates, irrigation frequency and trays counts, fully automating daily production planning.
- Implemented **JWT authentication** with **bcrypt** password hashing and expiring signed tokens, linking user accounts to employee records so requests resolve the acting employee and attribute orders to them.

Pokémon Battle Game | *TypeScript, React, Socket.IO, Express.js, Node.js* Jul 2025 – Sep 2025

- Built a **server-authoritative real-time multiplayer system** with Socket.IO and Express, using room-code lobbies and a typed event protocol that holds each turn until both players submit, then sends each client its own view of the battle state.
- Saved each server-resolved turn into an **ordered event log** (attack, switch, faint) that the React client plays back from a queue, driving Motion animations for attacks, switches, HP changes and faints so both players see the same sequence.

Achievements

2026 **Winner - Best Game**, GoatHacks 2026 - Worcester Polytechnic Institute *Worcester, MA*